

Homework for Chap 9

In a piece of semiconductor of GaAs, given the static dielectric constant is 12.9, the effective mass for electron is $0.067m_0$, the effective masses for heavy hole and light hole are $0.45m_0$ and $0.082 m_0$, respectively. Here m_0 is the rest mass of electron, and $m_0=9.11 \times 10^{-31}$ kg. Calculate the Bohr radius and binding energy of exciton ground state in GaAs. Please note that electron can form exciton with both heavy hole and light hole.